

Executive Summary

ATOMiK · State-Aware Compute Architecture · Pre-Seed · May 2026

What ATOMiK Is

ATOMiK is a state-aware compute architecture that helps constrained edge, embedded, and AI-at-the-edge teams evaluate whether they can reduce wasted state movement by tracking meaningful change instead of repeatedly moving, scanning, syncing, replaying, or rebuilding full state.

The company is raising a \$5M target pre-seed to move from hardware-backed primitive proof to measured customer workload proof, IP diligence, and ASIC feasibility. The round does not fund tape-out.

Why It Matters

- Buyer pain: battery budget, enclosure heat, link budget, update latency, reliability, size, and weight.
- First wedge: one state-heavy workload, one current baseline, and one painful constraint expensive enough to evaluate.
- Why now: energy, bandwidth, thermal, and local-compute pressure are rising; IEA/LBNL figures are market context, not ATOMiK savings claims.

Market Opportunity

TAM context	\$1T+ 2026 semiconductor revenue backdrop. Context only; ATOMiK does not address all semiconductor spend.
SAM context	\$112B-\$169B embedded systems market range from 2024 estimate to 2030 forecast; edge AI adds adjacent pressure.
Entry wedge	\$10M-\$40M early annual revenue path from paid evaluations, design partners, and licensing if proof converts; not a ceiling or forecast.
IP/licensing context	\$8.14B 2025 to \$11.2B 2029 semiconductor IP market context.

Evaluation Offer

Give us	One state-heavy workload, current baseline, and painful constraint.
We evaluate	Where state movement creates waste and whether ATOMiK can improve that path.
You receive	Workload map, baseline comparison, evidence map, fit/no-fit recommendation, and next-step plan.
Success	Measured improvement against one agreed metric while preserving correctness.

Proof Today

Artifact	Evidence Boundary
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Zynq Desk v0.40-A	HARDWARE_VALIDATED — live UI proof image captured from /dev/fb0, driven by real measured on-board data; not product maturity or performance proof.
Live Workloads demo	LIVE_MEASURED — first customer-facing workload demo on AX7020 hardware: 800 Mevents/s at 8 banks, byte-identical across configs. Throughput is measured; static capture, not interactive.
Parallel-bank throughput	LIVE_MEASURED — AX7020 8-bank accumulator scales 1/2/4/8x; byte-identical to software recompute. Throughput substrate, not a customer-workload speedup.
Linux userspace to FPGA	HARDWARE_VALIDATED — documented OS-to-bus path with 16/16 property checks.
AX7020 matrix	LIVE_MEASURED — workload-specific wins and losses; quote with caveats.
Lean4-checked formal algebra	FORMAL_PROOF where directly audited; otherwise SOFTWARE_VALIDATED / proof work present. Exact formal claims remain bounded to audited properties.
SD boot artifacts	BUILD_ARTIFACT — build output exists; power-on run artifact still gated.

Business Path

Paid proof reviews and technical evaluations -> design-partner evaluations -> IP/licensing diligence -> strategic licensing or platform partnership if workload proof supports it.

Defensibility wedge: Lean4-checked formal algebra, FPGA hardware validation, and an IP/proof registry. This is presented as a proof-bound diligence asset, not as a customer outcome claim.

Return narrative: pre-seed capital buys diligence assets a strategic partner can evaluate: workload proof, IP status, integration path, and ASIC feasibility. No exit, valuation, or return multiple is implied.

Ask

Target	\$5M pre-seed target; staged close terms counsel/CFO pending.
Use	\$1.5M engineering, \$1.0M customer proof, \$750K IP/legal, \$750K ASIC feasibility, \$1.0M finance/GTM/ops + reserve.
Need	Capital, design-partner introductions, qualified workload access, and ASIC/IP diligence support.
Pending	SAFE terms, valuation cap, discount, and close mechanics require CFO/counsel approval.

Evidence Boundary

Proof and financial claims remain artifact-bound or planning-only. Customer workload, production, revenue, and downstream battery, thermal, water, footprint, or ROI outcomes require matching measured artifacts or signed documents.

Contact

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